

IPD Cashless: Pre-Auth & Network Claims

Design Thinking-Led Product Case Study

Prepared for portfolio, interview, and leadership narrative use

Dimension	Summary
Organization / Context	Vidal Health TPA modernization under Bajaj Finserv Health
Project Coverage	IPD Cashless workflows covering Pre-Auth and Network Claims
Primary Systems / Product Areas	Vings legacy system, Enigma/DMS/Copilot, rule engine, queues, dashboards, insurer workflows
Primary Users	Processors, data entry, billing team, doctor team, TLs, auditors, provider/insurer operations
Core Objective	Reduce TAT, manual effort, rework, audit risk, and cost per claim while improving processing consistency and decision quality
Product Method	Design-thinking-led workflow discovery, decision decomposition, AI feasibility mapping, rule sequencing, queue design, and operational metric alignment

Executive Positioning

This case study is framed as an end-to-end IPD Cashless modernization initiative rather than a single Pre-Auth feature. The strongest narrative is the redesign of complex TPA operations into a structured, AI-assisted, rule-driven, audit-aware processing model spanning Pre-Auth and Network Claims.

Document Structure

- Executive summary and product context
- Design-thinking discovery and problem definition
- Pre-Auth workflow redesign and AI/rule-engine solutioning
- Network Claims workflow redesign and validation model
- Operating model, metrics, governance, and impact narrative
- Appendices: task breakdowns and interview-ready narrative

1. Executive Summary

The Vidal IPD Cashless modernization initiative focused on redesigning complex TPA workflows across Pre-Auth and Network Claims. These workflows were highly operational, document-heavy, rule-dependent, and time-sensitive. They involved multiple personas, fragmented system interactions, insurer dependencies, medical and billing validation, tariff/package checks, audit exposure, and strict turnaround expectations.

The product challenge was not merely to digitize an existing workflow. The real challenge was to decompose the claim journey into decision points, identify where manual effort and errors occurred, and design a processing model that could improve speed, accuracy, consistency, controllership, and scalability.

The solutioning approach combined design thinking, process decomposition, AI-agent opportunity mapping, rule-engine governance, document intelligence, queue prioritization, and metric-led execution. The resulting product narrative is best positioned as an IPD Cashless operating-system transformation: moving from fragmented manual processing toward an AI-assisted, rule-driven, evidence-backed, and auditable processing model.

2. Project Context

Area	Detail
Business Area	IPD Cashless processing within Vidal Health TPA modernization
Covered Workflows	Pre-Auth and Network Claims
Primary Business Problem	High manual effort and operational variability across authorization, medical validation, billing/tariff validation, documentation, concurrence, and claim decisioning
Primary Product Themes	Document intelligence, decision support, rule-engine sequencing, queue prioritization, AI-assisted processing, audit trail, dashboards, and insurer workflows
Critical Metrics	Fresh Pre-Auth TAT, Discharge TAT, AHT, FTR, STP, CPC, processor productivity, QA/audit error, rework, shortfall rate, and escalation rate
North Star Direction	Faster and more accurate cashless decisions with lower manual touchpoints and stronger auditability

Core Reframe

The project was not simply about automating claim processing. It was about redesigning the decision architecture of IPD Cashless operations: which decisions can be automated, which require human review, what evidence is needed, and how the system should preserve auditability.

3. Design Thinking Lens

For Vidal IPD Cashless, generic design-thinking methods were not enough. The correct approach was operational design thinking: observe the actual workflow, break it into tasks and decision points, classify leakage, design targeted interventions, and validate each intervention against measurable operational outcomes.

Design Thinking Stage	How It Applied to IPD Cashless
Empathize	Understand how processors, billing teams, doctors, TLs, auditors, and insurer teams actually work across systems, documents, rules, and escalations.
Define	Convert broad complaints such as high TAT into precise workflow problems: document completeness, tariff validation, rule sequencing, insurer concurrence, billing mismatch, or medical-code uncertainty.
Ideate	Explore multiple solution types: process redesign, queue strategy, document viewer, AI copilot, rule engine, dashboard, audit trail, insurer workflow, and exception handling.
Prototype / Model	Create granular task maps, AI-agent maps, rule sequencing models, queue logic, DMS flows, and processor decision-support concepts before scaling automation.
Test / Measure	Evaluate interventions using TAT, AHT, productivity, FTR,

	QA/audit error, CPC, rework, and user trust in system recommendations.
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4. User and Persona Map

Persona	Role in Workflow	Primary Pain / Need
Data Entry / Provider Claim Creation	Creates and updates cases, uploads or maps documents, initiates the case journey.	Needs clean intake, complete data, and reduced duplicate/manual entry.
TPA Medical Processor	Validates medical details, policy/rule applicability, documents, package/tariff, and decision status.	Needs one reliable view of documents, extracted facts, rules, confidence, and next action.
Billing Team	Reviews bills, packages, tariffs, non-payables, deductions, and final bill handling.	Needs accurate bill capture, package mapping, tariff logic, and rule-driven allowed amount calculation.
Doctor Team	Validates medical necessity, ICD/PCS, clinical details, and medical admissibility.	Needs concise clinical summary, document evidence, and exception visibility.
Team Lead / Manager	Manages queue, exceptions, OTP/rule-data review, escalation, and productivity.	Needs queue health, ageing, TAT breach, productivity, and exception controls.
Auditor / QA	Reviews decisions for correctness, auditability, and process adherence.	Needs decision trail, source evidence, override reason, and structured error taxonomy.
Insurer / Concurrence Team	Handles concurrence cases, insurer communication, and response tracking.	Needs structured handoff, email pack, status tracking, and MIS visibility.

5. Problem Definition

At a high level, Vidal IPD Cashless processing suffered from workflow fragmentation. Processors had to interpret documents, validate medical and billing information, apply rule logic, manage insurer dependencies, and make decisions under TAT pressure. The system needed to reduce manual effort without weakening medical judgment, financial controls, or auditability.

5.1 Core Problem Statement

Problem Statement

IPD Cashless processors needed a faster, more reliable, and audit-safe way to validate documents, medical information, billing/tariff details, policy rules, insurer concurrence, and final authorization decisions because the existing workflow required manual checks across fragmented systems and documents, resulting in higher TAT, rework, inconsistent deductions, audit risk, and high cost per claim.

5.2 Leakage Diagnosis

Leakage Type	Where It Appeared	Product Interpretation
Time Leakage	Document verification, tariff/package handling, final bill capture, insurer concurrence follow-up.	High AHT and queue ageing were not caused by one activity; they emerged across multiple micro-decisions.
Decision Leakage	Medical entry, ICD/PCS changes, package applicability, rule-data confidence, approval/shortfall/rejection.	Users needed stronger decision support and evidence rather than a simple automation button.
Data Leakage	Missing documents, incomplete PA forms, incorrect bill heads, wrong room category, inconsistent rule data.	Data needed to be extracted, validated, and surfaced before decisioning.
Trust Leakage	AI/rule suggestions without source evidence or confidence.	Recommendations required evidence links, confidence bands, and override controls.
Control Leakage	Manual deductions, early co-pay/ailment-cap application, insurer concurrence loops, OTP/rule-data exceptions.	Rule engine and audit trail needed to become the single point of truth.
Quality Leakage	Wrong tariff, wrong bill amount, incorrect room category, missed non-payables, wrong medical coding.	Quality controls needed to sit inside the workflow, not only after-the-fact audit.
Cost Leakage	High processor effort, repeated verification, manual insurer follow-up, rework.	Automation and assistive workflows needed to reduce CPC without compromising accuracy.

6. Pre-Auth Workflow Redesign

Pre-Auth was the most time-sensitive part of IPD Cashless. It covered fresh authorization, interim enhancement, and final discharge-related processing. The product design had to handle document completeness, medical validation, tariff/package logic, rule-data confidence, insurer concurrence, and final authorization actions.

6.1 Pre-Auth Stages

Stage	Workflow Focus	Product Opportunity
Fresh Pre-Auth	Initial authorization request, document verification, medical entry, rule data, tariff/package/open bill handling, decision save.	Early document completeness, medical summary, rule confidence, and tariff/package assistance.
Interim Request	Additional authorization during hospitalization; compare current papers with prior requests and documents.	Delta detection, prior-request comparison, document continuity, and enhancement logic.
Final Request / Discharge	Final bill capture, discharge summary review, final tariff/package/open bill validation, authorization save.	Final bill digitization, non-payable detection, deduction sequencing, and discharge TAT control.
Insurer Concurrence	Email pack creation, revert tracking, MIS, and insurer response dependency.	Concurrence tracker, auto-mailers, response parsing, SLA visibility, and status dashboards.

6.2 Pre-Auth Solution Tracks

Solution Track	Purpose	Representative Capabilities
Queue Strategy	Prioritize cases based on urgency, risk, specialty, and TAT.	Discharge Fast-Lane, Pre-Auth Routine, Pre-Auth High-Risk, Specialty queues, Emergency, Appeals/Exception.
Document Intelligence	Reduce manual document search and verification effort.	Document completeness, chronological viewer, right-side DMS, compare view, audit view, missing-document flags.
Medical Decision Support	Support clinical validation without replacing medical judgment.	Clinical summary, ICD/PCS suggestions, medical entry validation, exception routing for doctor review.
Tariff / Package Assistant	Reduce manual tariff/package computation and errors.	Room-linked package detection, caps, NTBC, package deviation handling, bill comparison.
Rule Engine Governance	Make rule application consistent and auditable.	Single source of truth for deduction sequence, confidence checks, OTP/TL exception workflow.
Final Bill Support	Reduce final discharge processing time and missed deductions.	Final bill capture, non-payable checks, allowed amount computation, discharge TAT support.
Insurer Workflow	Reduce manual concurrence follow-up and MIS effort.	Auto-mailer, email pack, concurrence tracker, response capture, SLA alerts.
Dashboards	Provide operating visibility to TLs and managers.	TAT, queue ageing, productivity, decisions, QA errors, audit errors, CPC, shortfall/rework.

6.3 Rule Engine and Deduction Sequencing

One of the most important product principles was to prevent manual deductions from bypassing the rule engine. In the intended model, the rule engine becomes the single point of truth for final allowed amount calculation, while UI and backend controls restrict premature manual deduction inputs.

Deduction Sequence

Hospital Discount -> Package Price -> NME -> Room Rent Limit -> Co-pay -> Ailment Cap -> Threshold -> Balance Sum Insured

This sequence matters because applying co-pay, ailment cap, or other deductions too early can distort the final approved amount and create inconsistent outcomes. The design direction was to enforce sequencing through backend validations, UI restrictions, and rule-engine-driven final amount calculation.

7. Network Claims Workflow Redesign

Network Claims was adjacent to Pre-Auth but had a different operating profile. It involved planning the day, opening allocated claims, comparing documents and case summary with PA forms, validating medical details, checking rejection/shortfall pointers, billing/tariff validation, rule-data checks, approval letter generation, and insurer concurrence.

7.1 Network Claims Problem Statement

Problem Statement

Network Claims processors needed structured validation of document, billing, tariff, rule, and medical details because manual reconciliation and incorrect extraction could lead to wrong approvals, delays, rework, and insurer/customer escalation.

7.2 Network Claims Solution Tracks

Solution Track	Purpose	Representative Capabilities
Planning and Allocation	Help processors plan the day and work from a clean claim list.	Auto allocation, ageing visibility, daily worklist, complexity/risk tagging.
Document and Medical Review	Compare IC papers, case summary, PA form, previous docs, and latest uploads.	Document compare, medical detail validation, ICD update routing.
Rejection / Shortfall Decision Support	Identify query/rejection pointers and policy/episode checks.	Policy sanity, data sanity, medical episode checks, structured reason capture.
Billing and Tariff Validation	Reduce wrong bill/tariff capture and room category errors.	Bill head capture, tariff mapping, room category validation, implant invoice detection.
Rule Data Checker	Ensure rule checks are run and captured consistently.	Checker flow, confidence flags, exceptions.
Claim Letter Generation	Standardize approval output and remarks.	Approval letter generation, authorization type, final remarks.
Insurer Concurrence	Move manual follow-up into structured tracking.	Concurrence response capture, MIS, status tracking, partial automation.

8. AI-Agent and Copilot Opportunity Map

The initiative should be positioned as AI-assisted transformation rather than blind automation. The right automation design was to identify repeatable, evidence-backed tasks suitable for AI assistance while preserving human review for judgment-heavy or audit-sensitive decisions.

Agent / Capability	Primary User	Purpose	AI Need / Why
Document Completeness Agent	Processor / Data Entry	Detect missing or unreadable documents before detailed processing.	Reduces time spent opening every file and prevents late shortfalls.
Medical Summary Assistant	Processor / Doctor Team	Summarize clinical documents, diagnosis, procedure, and relevant history.	Improves speed while preserving doctor review for medical judgment.
ICD/PCS Support	Processor / Doctor Team	Suggest or validate medical coding changes.	Reduces coding effort and routes uncertain cases to medical review.
Tariff / Package Assistant	Billing / Processor	Map package, tariff, room category, caps, deviations, and NTBC logic.	Targets a high-error, high-effort activity.
Open Bill Analyzer	Billing / Processor	Compute bill heads, non-payables, and final allowed amount support.	Reduces final-stage manual calculation effort.
Rule Engine Sequencer	Processor / TL	Apply deductions in approved order and prevent premature manual inputs.	Improves consistency, auditability, and financial correctness.
Concurrence Tracker	Processor / Insurer Desk	Track insurer cases, responses, status, and SLA.	Reduces manual follow-up and TAT leakage.
Audit Risk Assistant	Auditor / TL	Flag cases likely to need QA review based on rule exceptions, overrides, or evidence gaps.	Improves controllership and targeted sampling.

9. Metrics and Measurement Model

The initiative was designed around operational performance, not feature completion alone. The metric model should be presented at three levels: speed, quality, and cost/controllership.

Metric Category	Metrics	Why It Matters
Speed	Fresh Pre-Auth TAT < 60 minutes, Discharge TAT < 3 hours, AHT, queue ageing.	Measures whether the workflow improves time-sensitive authorization and discharge handling.
Productivity	Transactions per processor/day, decisions per processor/day, processing capacity.	Measures scalability and manual effort reduction.
Quality	FTR, shortfall rate, rework rate, wrong deduction, wrong tariff/billing capture.	Measures decision correctness and reduction in avoidable rework.
Automation	STP %, AI-assisted completion, manual touchpoints per case, override rate.	Measures movement from manual processing to assisted or straight-through processing.
Controllership	QA error %, audit error %, rule-engine bypass, override reason completeness.	Measures auditability and process governance.
Cost	Cost per claim / cost per case, processor capacity, manual effort reduction.	Measures business value and operating leverage.

10. Governance and Risk Controls

Healthcare claims automation requires stronger governance than ordinary workflow automation. The design needed to preserve auditability, override control, evidence traceability, and human accountability.

- Human-in-the-loop design for judgment-heavy medical and financial decisions.
- Evidence-linked recommendations for AI/copilot outputs.
- Confidence flags and exception routing for low-confidence outputs.
- Role-based access for processors, TLs, doctors, auditors, and managers.
- Audit trail for rule application, manual override, decision change, and final authorization.
- Backend validations to prevent rule-engine bypass or incorrect deduction sequencing.
- Structured reason codes for shortfall, rejection, approval, concurrence, and override actions.

11. Business Impact Narrative

The strongest credible impact narrative is that the initiative created a structured modernization blueprint for IPD Cashless operations, translating fragmented manual workflows into a rule-driven, AI-assisted, and auditable operating model across Pre-Auth and Network Claims.

The value areas were:

- Reduced TAT risk through queue prioritization, document intelligence, and faster decision support.
- Reduced processor effort by decomposing workflow into AI-assistable tasks.
- Improved decision consistency through rule-engine sequencing and structured validations.
- Improved auditability through evidence links, decision trails, and override capture.
- Improved scalability by moving from manual checks toward assisted processing and STP-ready capabilities.
- Improved controllership by preventing premature/manual deduction bypass and enforcing rule order.

12. Your Role and Leadership Narrative

As product lead for the Vidal IPD Cashless modernization effort, the role should be positioned around workflow discovery, problem decomposition, AI-agent mapping, rule-engine governance, stakeholder alignment, and metric-led transformation.

Responsibility Area	Contribution
Workflow Discovery	Mapped granular Pre-Auth and Network Claims activities, personas, handoffs, dependencies, and decision points.
Product Solutioning	Converted operational pain points into solution tracks across DMS, copilot, rule engine, queue management, dashboards, and insurer workflows.

AI Transformation	Identified tasks suitable for AI assistance versus human review; framed agentic opportunities across document, medical, tariff, billing, and audit workflows.
Rule Engine Governance	Defined sequencing logic and controls to make the rule engine the single source of truth for deductions and final allowed amount.
Metrics Alignment	Aligned the transformation with TAT, AHT, FTR, STP, CPC, QA/audit error, productivity, and rework metrics.
Stakeholder Management	Worked across operations, processors, TLs, doctor team, billing, QA/audit, engineering, and insurer-related workflows.

13. Interview-Ready Narrative

Narrative

At Vidal Health, I worked on IPD Cashless modernization covering Pre-Auth and Network Claims. The challenge was not just digitizing a workflow; it was redesigning a complex TPA operating model involving documents, medical validation, billing, tariffs, policy rules, insurer concurrence, discharge urgency, and audit controls.

I approached the problem through operational design thinking. We decomposed the journey into granular tasks, identified where time, data, decision, quality, and control leakage happened, and mapped which activities could be assisted through AI, rule engines, document intelligence, and queueing.

For Pre-Auth, the focus was on fresh, interim, and final authorization flows, including document verification, medical entry, tariff/package handling, rule-data checks, final bill validation, and insurer concurrence. For Network Claims, the focus was on document and medical validation, billing/tariff accuracy, rule checks, approval letter generation, and concurrence support.

A critical part of the solution was rule-engine governance. We defined a structured deduction sequence and positioned the rule engine as the single source of truth to prevent premature manual deductions and improve auditability. The broader product direction was to move Vidal from fragmented manual processing toward an AI-assisted, evidence-backed, and auditable IPD Cashless operating model.

Appendix A: Pre-Auth Granular Task Breakdown

This task map is used to show how the workflow was decomposed for solutioning and AI-assist mapping. Times are in minutes where known.

#	Pre-Auth Task	Time	Agentic Scope
1	Submit pre-auth (email inwarding)	-	No
2	Document upload (estimate, ID, clinicals)	1.0	No
3	Provider portal: information fill and document upload (auto inwarded)	2.0	Yes
4	Document verification (policy BSI/expiry; proceed or shortfall)	2.0	Yes
5	Medical entry (validate helpdesk details, add missing, ICD/PCS)	-	Partially Yes
6	Rule data read and confidence check; TL OTP / Rules team if incorrect	0.3	No
7	Tariff/Package handling (room-linked, caps, NTBC, compute and submit)	8	Yes
8	Open Bill (no package): fetch requested from PA form	-	Yes
9	Authorization and decision save (type/status/reason/remarks)	1	No
10	Insurer concurrence (email pack, track revert, MIS)	20	No
11	Submit interim request (email inwarding; Fresh to Interim etc.)	2	No
12	Interim - Document verification (compare IC papers, PA form, prior reqs)	2	Yes
13	Interim - Medical entry (validate/edit; move to resolution if ICD/PCS updates)	2	Partially Yes
14	Interim - Rule data and OTP (confidence,	1	No

	TL OTP; Rules team)		
15	Interim - Tariff/Package (deviation handling, NTBC, approve and submit)	2	Yes
16	Interim - Open Bill (no package): fetch from interim bill	5	No
17	Interim - Authorization and save (trigger letters)	2	No
18	Interim - Email concurrence (loop + MIS)	20	No
19	Submit final request (email inwarding; Interim to Final etc.)	5	No
20	Final bill capture (manual): reconcile and enter mapped headers to Vings	10	Yes
21	Final - Document verification (discharge summary + prior docs; proceed/shortfall)	2	Yes
22	Final - Medical entry (validate/edit; move to resolution if ICD/PCS)	2	Yes
23	Final - Rule data and OTP (TL OTP; Rules team if incorrect)	0.25	No
24	Final - Tariff/Package (repeat flow; deviation handling)	2	Yes
25	Final - Open Bill (no package): compute post non-payables; approve final	5	Yes
26	Final - Authorization and save (trigger letters; queue exit)	2	No
27	Final - Email concurrence (loop + MIS)	20	No

Appendix B: Network Claims Granular Task Breakdown

#	Network Claims Task	Time	Agentic Scope
1	Pre-Requisites: Manual inwards done; manager informed of dump; auto allocation to processor	NA	NA
2	Planning the day: get list of Netclaims, plan day	10	YES
3	General and Medical: open case; compare IC papers/case summary with PA form; validate/edit medical details; move to resolution if ICD updates; view latest and previous docs; proceed if complete else raise shortfall	5	YES
4	Rejection/Shortfalls Decisions: check pointers for query/rejection; review policy rule check, data sanity check, medical episode check	2	YES
5	Billing and Tariff: billing of medical episode, log heads/subheads/amounts	3	YES
6	Rule Data: run rule data checks (checker flow)	1	YES
7	Claim Letter Generation: generate approval letter; fill remarks and final authorization type	2	YES
8	Insurer Concurrence: current flow largely manual; partial automation till response	2	YES

Appendix C: Portfolio Summary and Resume Bullet

Portfolio Summary:

Led Vidal IPD Cashless modernization solutioning across Pre-Auth and Network Claims by decomposing complex TPA workflows into granular decision points, mapping AI-agent opportunities, defining rule-engine sequencing, and designing DMS, queue, dashboard, insurer workflow, and audit-control capabilities to improve TAT, productivity, accuracy, controllership, and cost per claim.

Resume Bullet:

Led IPD Cashless modernization for Vidal Health across Pre-Auth and Network Claims, designing AI-assisted workflows, document intelligence, rule-engine deduction sequencing, queue prioritization, tariff/billing validation, insurer concurrence tracking, dashboards, and audit controls to reduce manual effort, improve TAT, and strengthen claims controllership.